YieldLedger: separating plastic dissipation from endpoint-work error

Computational mechanics project report

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Abstract

An energy balance can close while its interpretation remains wrong: in incremental plasticity, endpoint force–displacement work includes a nonnegative quadrature excess that is distinct from irreversible material dissipation. This report studies that distinction in a small-strain uniaxial mixed-hardening model. An original C++ return-mapping kernel is coupled to a Python bar finite-element solver with committed material history and a safeguarded Newton iteration. Synthetic cyclic and analytical bar experiments verify yield admissibility, consistent tangents and a discrete energy identity. The final work-excess refinement order is 0.998363. For a two-element bar, consistent and elastic tangents require one and 72 corrections, respectively. These results verify a specified numerical model; they do not validate a physical material or establish a general solver speedup.

1 Question, assumptions and constitutive model

The research question is whether plastic dissipation can be reported independently of load-increment work error while preserving a reliable global equilibrium solve. Return mapping and consistent tangents are standard tools in computational plasticity [1]; mixed hardening provides both a translating and expanding yield interval [2]. Here their simplest one-dimensional linear form makes the energy accounting and reference solutions transparent. No new constitutive law or return-mapping algorithm is claimed.

Consider a bar with constant area A, no body force, imposed end displacements $u(0) = 0$, $u(L) = U$, and infinitesimal axial strain $\varepsilon = u'$. Quasistatic equilibrium is $(A\sigma)' = 0$. The internal variables are signed plastic strain p and accumulated plastic strain $\alpha \geq |p|$. With positive elastic modulus E and initial yield stress σ_y, and nonnegative hardening moduli H_i, H_k,

$$\sigma = E(\varepsilon - p), \qquad f = |\sigma - H_k p| - (\sigma_y + H_i \alpha) \leq 0, \tag{1}$$

$$\mathrm{d}p = s\,\mathrm{d}\gamma, \qquad \mathrm{d}\alpha = \mathrm{d}\gamma \geq 0, \quad f\,\mathrm{d}\gamma = 0, \quad s = \operatorname{sign}(\sigma - H_k p). \tag{2}$$

The model is rate-independent and isothermal. Hardening contributions are stored in the chosen free-energy function, not counted as irreversible dissipation. Stored isotropic hardening energy is not recovered by elastic unloading because α cannot decrease. This constitutive assumption is not inferred from tests.

2 Incremental algorithm and energy identity

At each material point, the old state (p_n, α_n) is held fixed during the global solve. Given trial strain ε_{n+1}, define

$$\sigma^{\mathrm{tr}} = E(\varepsilon_{n+1} - p_n), \qquad \xi^{\mathrm{tr}} = \sigma^{\mathrm{tr}} - H_k p_n, \tag{3}$$

$$f^{\mathrm{tr}} = |\xi^{\mathrm{tr}}| - \sigma_y - H_i \alpha_n, \qquad \Delta\gamma = \frac{\max(0, f^{\mathrm{tr}})}{E + H_i + H_k}. \tag{4}$$

Set $s = \mathrm{sign}(\xi^{\mathrm{tr}})$, $p_{n+1} = p_n + s\Delta\gamma$, $\alpha_{n+1} = \alpha_n + \Delta\gamma$, and $\sigma_{n+1} = \sigma^{\mathrm{tr}} - Es\Delta\gamma$. The algorithmic tangent is E on the elastic branch and $E(H_i+H_k)/(E+H_i+H_k)$ on the plastic branch. At the yield kink the code selects the elastic one-sided tangent; finite-difference derivative checks deliberately avoid that point.

With $e = \varepsilon - p$, the stored energy and incremental irreversible dissipation are

$$\psi = \tfrac{1}{2}Ee^2 + \tfrac{1}{2}H_i\alpha^2 + \tfrac{1}{2}H_k p^2, \qquad \Delta D = \sigma_y \Delta\alpha. \tag{5}$$

Using $x_{n+1}\Delta x = \Delta(x^2)/2 + (\Delta x)^2/2$ and the return condition gives

$$\sigma_{n+1}\Delta\varepsilon = \Delta\psi + \Delta D + Q, \tag{6}$$

$$Q = \tfrac{1}{2}E(\Delta e)^2 + \tfrac{1}{2}H_i(\Delta\alpha)^2 + \tfrac{1}{2}H_k(\Delta p)^2 \geq 0. \tag{7}$$

In plastic loading, $\sigma_{n+1}\Delta p = (\sigma_y + H_i\alpha_{n+1})\Delta\alpha + H_k p_{n+1}\Delta p$; in elasticity both plastic increments vanish. Thus the identity also holds for steps that cross yield. Summing it separates endpoint work, stored energy, irreversible dissipation and Q without an inferred residual category.

2.1 Implementation and global solution

The C++17 header implements the closed-form update, finite-input/state checks and stored energy. pybind11 exposes read-only material, state and response objects. Python assembles P1 bar elements with one constant-strain point each: $\mathbf{f}_e = A\sigma_e[-1, 1]^T$ and $\mathbf{K}_e = AC_e \left[\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right] / \ell_e$. The free-node residual is solved with dense linear algebra. A correction is halved up to 29 times until the infinity-norm residual decreases by an Armijo factor 10^{-4} or meets the final force tolerance. Each trial starts from the same old material state. Only convergence commits history; failed solves raise an exception. Positive total hardening is required by the bar API to exclude fully plastic singular tangents. The point kernel separately supports perfect plasticity.

2.2 Reproducible experiment definitions

Material-point stresses use MPa: $E = 200000$, $\sigma_y = 250$, $H_i = 2000$, $H_k = 3000$. Strain visits zero, 0.008, -0.008, 0.008, and zero, with N equal increments per leg, $N = 25, 50, 100, 200, 400, 800$. Every reversal is sampled; $N = 400$ is the main record. A 501-point monotonic path from zero to 0.02 is compared with the exact bilinear law. Four tangent checks use central differences of size 10^{-8} after preloading to 0.004. Their error is normalized by E.

3 Measured validation results

The bar has $A = 10\,\mathrm{mm}^2$, layer lengths $(40, 60)$ mm, elastic moduli $(200000, 100000)$ MPa, yield stresses $(200, 400)$ MPa, and total hardening moduli $(6000, 10000)$ MPa. Their (H_i, H_k) splits are $(2000, 4000)$ and $(5000, 5000)$. At $U = 0.4$ mm, only the first layer is plastic. Equilibrium and compatibility give

$$\sigma_* = \frac{U + L_1\sigma_{y1}/H_1}{L_1/E_1 + L_2/E_2 + L_1/H_1} = 232.142857 \text{ MPa}. \tag{8}$$

Meshes split each layer into 1, 4 or 16 elements. Newton stopping uses absolute tolerance 10^{-10} N plus $10^{-10}\max(1, \max_e |A\sigma_e|)$ N. The iteration comparison starts from a virgin two-element bar and an affine displacement guess.

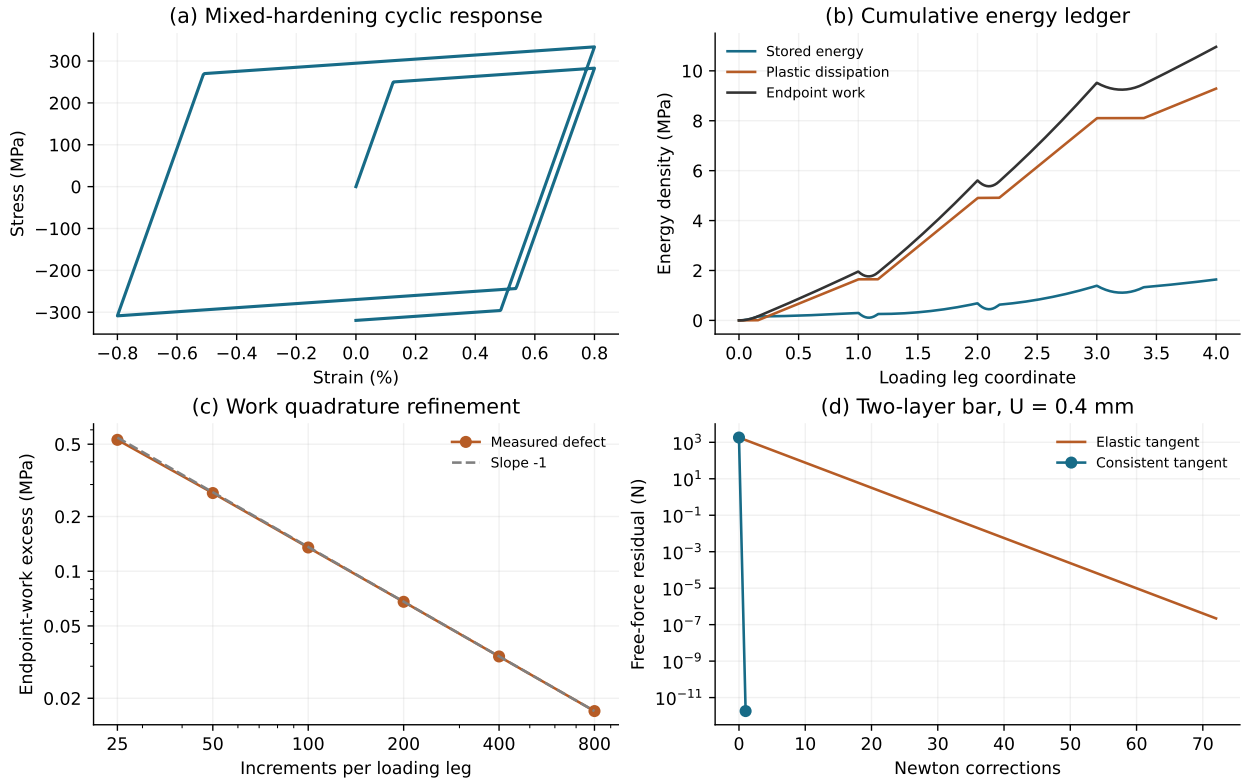


Figure 1: Original recorded experiments: cyclic stress response; cumulative energy components; endpoint-work excess under refinement; and the two-element bar residual. The last panel compares corrections, not execution time.

Verification quantity	Recorded value
Maximum monotonic stress error (MPa)	5.684×10^{-14}
Maximum tangent error divided by E	9.093×10^{-11}
Maximum cumulative energy-identity error (MPa)	8.704×10^{-14}
Maximum aligned-bar stress error, all meshes (MPa)	1.364×10^{-12}
Maximum aligned-bar nodal error, all meshes (mm)	3.331×10^{-16}
Consistent / elastic tangent corrections, two elements	1 / 72

Table 1: Deterministic verification values from the committed numerical outputs. Near-roundoff errors are platform dependent, not attainable accuracy guarantees.

4 Discussion: what the errors do and do not measure

For the main 1,600-increment path, final stored energy is 1.639064 MPa, irreversible dissipation is 9.287179 MPa, endpoint work is 10.960207 MPa, and endpoint-work excess is 0.033964 MPa. Here MPa is an energy density ($\mathrm{N\,mm/mm^3}$). The final strain is zero but internal variables and residual stress need not vanish. Consequently this is not a closed state-space cycle and its total work must not be equated with plastic dissipation alone.

For this one-dimensional law, return mapping gives exact endpoint states along each monotonic imposed-strain leg, including a yield transition. Continuum work on the resolved path is the change in stored energy plus $\sigma_y \Delta\alpha$. Thus Q is a work-quadrature error, not an added physical mechanism. Its measured order approaches one: the final pair of refinements gives 0.998363. This observation does not establish first-order error for every internal variable or every loading path. Reversals absent from the sampled path cannot be reconstructed by this method.

The aligned bar has piecewise constant strain and piecewise affine displacement. The exact continuum solution therefore lies in every tested P1 space. Table 1 primarily measures nonlinear algebraic and floating-point error, not a spatial convergence rate. No general claim is made about curved geometries, non-aligned interfaces, heterogeneous elements, or multiaxial finite elements. The one-versus-72 comparison demonstrates the value of the consistent derivative for this particular equilibrium problem. It does not measure runtime, account for all trial evaluations, or prove global Newton convergence. Backtracking was needed to prevent active-set cycling in refined meshes whose initial guesses lay at yield.

5 Reproducibility, limits and conclusion

The repository includes 39 Python tests and a native 2,000-update cyclic invariant test covering analytical response, reverse yield, complementarity, tangents, energy, equilibrium and failed-step state preservation. Run `yieldledger -output results` and `python paper/build.py -engine pdflatex` from an installed checkout. The builder derives numerical macros from recorded metrics, hashes inputs, checks references and layout, and enforces a five-page limit. CSV data and CI are included.

No physical calibration is provided. Finite strains, thermal effects, viscosity, damage, multiaxial response and fatigue are excluded. Unsaturated linear hardening is an idealization. Dense matrices, no automatic step cutback, and bounded backtracking limit scale and robustness. Numerical verification does not establish physical validity or suitability for design.

Within these limits, the explicit ledger separates dissipation from endpoint-work excess, while committed-state updates and consistent tangents produce a verifiable equilibrium solution. Separate reporting makes refinement results interpretable.

References

[1] J. Bleyer et al., *Elasto-plastic analysis of a 2D von Mises material*, Computational Mechanics Numerical Tours with FEniCSx. Public numerical-method background. `https://bleyerj.github.io/comet-fenicsx/tours/nonlinear_problems/plasticity/plasticity.html`. Accessed 30 September 2026.

[2] MOOSE, *Combined Nonlinear Hardening Plasticity*, public framework documentation. Describes broader nonlinear hardening models than used here. `https://mooseframework.inl.gov/moose/source/materials/CombinedNonlinearHardeningPlasticity.html`. Accessed 30 September 2026.